

License Plate Detection

ITAI 1378 - Computer Vision and AI
Midterm Project Proposal

Tier 1 - Core

Chris Roy
Njeh Ababio
Unnati Shakya
Edwin Marquez



The Problem

Manually identifying license plates in large vehicle image datasets is **slow and inefficient**.

Parking system operators, traffic monitoring systems, and automated vehicle-management systems rely on it.

Automating license plate detection delivers **faster** processing, **consistent** results, and **scalable** operations.



Our Solution

A computer vision application that automatically detects license plates using a YOLO object detection model.



01

Vehicle Image



YOLO

Object Detection
Model



03

Detected License Plate
(Annotated Output)

02

YOLO Object
Detection Model

Computer Vision Technique

Object Detection with **YOLO**

- Single regression
- Fast inference speeds
- Low false-positive predictions

Ultralytics YOLO with **PyTorch**

- Dynamic Computational Graph
- Seamless Integration



Data Plan

License Plate Dataset by Ronak Gohil on Kaggle, 1,695 vehicle images.

<https://www.kaggle.com/datasets/ronakgohil/license-plate-dataset>

One class: license_plate; bounding-box annotations are in YOLO format.

Images are reviewed for coordinate accuracy and normalized to **640 x 640 resolution**.

Reviewed for errors, then **split** into 80% training, 10% validation, and 10% test sets

A **pre-annotated dataset** minimizes data preparation and fits the Tier 1 scope timeline



Success Metrics

Detection accuracy

mAP@50 ≥ 0.70

measures bounding-box detection accuracy

Precision ≥ 0.80

measures correctness of detected license plates

Inference Latency (≤ 100 ms per image)

measures the total time delay between input and output



Milestone Plan

01	Submit the proposal , outline project architecture, and create the GitHub repository	Week 8
02	Run a pretrained YOLO model end to end on a few license-plate images	Week 9
03	Prepare the Kaggle dataset, validate raw images, and produce annotated output images	Week 10
04	Fine-tune YOLO, training curves, and implement bounding box cropping pipeline	Week 11 & 12
05	Testing on the held-out test set, record mAP@50, precision, and inference time	Week 13 & 14
06	Record demo, complete README, finalize presentation slides, and submit Final	Week 15

Risks & Resources

01

Dataset Quality

inconsistent, missing, or
hard-to-detect plates



Plan B

clean annotations,
apply data augmentation,
or switch to alternate
public dataset

02

Model Performance

issues with lighting,
angles, distances,
occlusions



Plan B

further fine-tuning,
adjust training parameters,
limit scope to clear
plate images

Compute & Cost

Google Colab free GPU resources

Tools

Ultralytics YOLO +
PyTorch (open source)

Expected project cost

\$0